

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

User-flagged MODERATE risk. The user accepted the eight categories as necessary and sufficient [A3], reducing one source of ontological gap. However, the benchmark's output taxonomy operationalizes Equitable Tutoring [Q97, Q98, Q166] with an explicitly anti-telling, constructivist orientation, treating Telling as a last resort [Q155] and using Telling@k as a negative metric [Q96] — directly conflicting with the Indian metro pedagogical norm of direct correction as a primary, not last-resort, strategy. The user identified 'providing guidance' as the single most critical dimension [A3], but the benchmark distributes evaluation across three co-equal dimensions (Coherence, Correctness, Equitable Tutoring) [Q97, Q166], not weighting guidance commensurately. Dataset analysis confirms structural embedding of these values: Telling is positioned as late-stage fallback after multiple Focus/Probing failures [DATASET-D3, DATASET-D5, DATASET-D27], and self-correctness annotations frame Telling as 'had to reveal' [DATASET-D5, DATASET-D23] — a quality-deficit framing rather than legitimate pedagogical strategy. Additionally, the benchmark's classification of teacher openings that fully restate correct setups as 'Generic' rather than 'Telling' [DATASET-D12, DATASET-D28] reveals taxonomy ambiguity Indian annotators might resolve differently. This is structural validity violation: the construct's structure (what counts as good guidance) is misrepresented for the target context.

ID	Evidence
Q96	"Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)
Q97	"Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)
Q98	"Equitable tutoring describes how well the model provides the student with room for exploring the problem and solution space." (p.8)
Q155	"On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17)
Q166	"The following dimensions were rated by annotators: Coherence - 'The response naturally follows up on the previous utterance and context and has no logical conflicts with the context.' Correctness - 'The response is factually and mathematically correct and respects the learning concepts being taught.' Equitable tutoring - 'The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways.'" (p.18)
D3	Carmela has $600 \times 2 + 50 = 1250$... "It is 1250. Not 2450" ... $600 + 950 + 1250 = 2800$ — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback
D5	No $32 - 7 = 25$. So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation
D12	Teacher: Hi Riya. Shawn painted 9 red pebbles, 13 blue pebbles, and the 18 remaining pebbles into 3 equal groups of 6 yellow, 6 purple, and 6 green pebbles. Correct? — Teacher opens by recapping the correct setup — essentially pre-solving before asking student; minimal scaffolding effort
D23	Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =" / "Teacher: (telling)4 is the correct answer you don't need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction
D27	Teacher: (telling)19 - 14 = 5 minutes. — Single arithmetic Telling move that resolves the conversation; teacher had already spent multiple Focus moves before telling

D28	Teacher: (generic)Hi Riya. Shawn painted 9 red pebbles, 13 blue pebbles, and the 18 remaining pebbles into 3 equal groups of 6 yellow, 6 purple, and 6 green pebbles. Correct? — Opening move presents the full correct setup to student before posing the actual question; near-complete Telling at the outset
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Strengths

- Eight pedagogical categories user-confirmed as necessary and sufficient [A3]
- Explicit operational definitions provided for each output dimension [Q166]
- Three-dimensional human evaluation framework offers some richness [Q97]
- Self-correctness annotations enable filtering by tutoring strategy used [DATASET-D5, DATASET-D23]

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Q97	"Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)
Q166	"The following dimensions were rated by annotators: Coherence - "The response naturally follows up on the previous utterance and context and has no logical conflicts with the context." Correctness - "The response is factually and mathematically correct and respects the learning concepts being taught." Equitable tutoring - "The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways."" (p.18)
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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are user-accepted as relevant [A3], but the Equitable Tutoring dimension [Q98, Q166] encodes a constructivist anti-telling orientation that does not match Indian metro pedagogical priorities [A4].

OO-2: No new categories required per user [A3], but the deployment requires that 'providing guidance' be weighted as primary [A3], whereas the benchmark treats it as one of three co-equal dimensions [Q97].

OO-3: Yes — Equitable Tutoring [Q98, Q166] explicitly rewards withholding answers and giving room for exploration, encoding constructivist Western values misaligned with Indian direct-correction norms. Telling@k [Q96] frames direct instruction as a failure mode.

OO-4: Stakeholder-driven taxonomy reweighting (not redesign) is warranted: 'providing guidance' should be elevated as primary and Telling reframed as legitimate strategy rather than last-resort failure [A3, A4, DATASET-D5, DATASET-D23].

OO-5: Structural validity issue documented: the benchmark's scoring function encodes a Socratic anti-telling preference [Q154, Q155, Q96] that the user explicitly identifies as misaligned with Indian professional teacher pedagogy [A4].

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Q96	"Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)
Q97	"Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)
Q98	"Equitable tutoring describes how well the model provides the student with room for exploring the problem and solution space." (p.8)
Q154	"Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing." (p.17)
Q155	"On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17)
Q166	"The following dimensions were rated by annotators: Coherence - "The response naturally follows up on the previous utterance and context and has no logical conflicts with the context." Correctness - "The response is factually and mathematically correct and respects the learning concepts being taught." Equitable tutoring - "The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways."" (p.18)
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D23	Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =?" / "Teacher: (telling)4 is the correct answer you don't need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction

Information Gaps

- Whether the 'providing guidance' dimension the user prioritized [A3] is operationally equivalent to the benchmark's Coherence+Correctness combination, or to Equitable Tutoring, or to none of these

Requires Expert Verification

- Whether Indian metro teachers would re-classify dialogues like DATASET-D12 and DATASET-D28 (Generic openings that pre-solve) as Telling — testing operational stability of the taxonomy across populations
- Optimal reweighting of the three human-evaluation dimensions for Indian pedagogical priorities

Remediation

Gap 1: Telling@k frames direct instruction as a failure mode and Equitable Tutoring rewards withholding answers — both encode constructivist values misaligned with Indian professional teacher priorities. 'Providing guidance' (user-confirmed primary dimension) is not commensurately weighted.

Recommendation: Replace Telling@k with a 'Guidance Adequacy@k' metric calibrated by Indian teacher judgments of when directive telling is appropriate. Reweight the three human-evaluation dimensions (Coherence, Correctness, Equitable Tutoring) to give 'Providing Guidance' (operationalized as adequacy and timeliness of directive instruction) primary weight per user elicitation.